

Data for Policy Analysis and Management

CLASS 9: EXPERIMENTAL AND
QUASI-EXPERIMENTAL
METHODS CONT.

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Summer 2026

Agenda

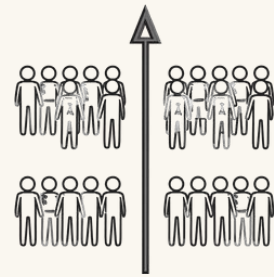
1. RCTs
2. Experimental Design
3. Internal Validity
4. Critiques
5. HOPE VI Example
6. Lab: Working on your final project

Randomized Controlled Trials: Strengths, Limitations, and Alternatives

Are randomized controlled trials always the best way to understand social and policy interventions?

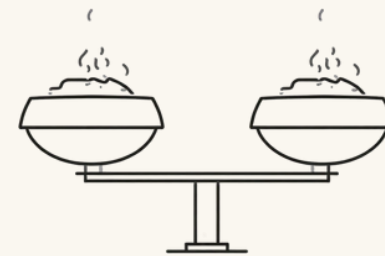


Why Researchers Value Randomized Controlled Trials



Main Advantage

- Randomization solves the fundamental problem of causal inference:
 - *What would have happened if the treated group had not received the intervention?*



Key Point

- Random assignment creates:
 - Treatment group
 - Control group
- Both are expected to be equivalent before treatment.



Outcome

- Any differences observed after treatment **can be confidently attributed to the intervention**, not to pre-existing group differences.

Internal Validity: The Strength of RCTs

Why RCTs Are the Gold Standard

- Randomization creates treatment and control groups that are statistically equivalent before the intervention.
- Selection bias is eliminated → motivation and background are equal across groups.
- Confounding variables are balanced between treatment and control groups.
- Alternative explanations are isolated, strengthening causal claims.

A randomized evaluation of a job training program:

- Without randomization: Participants may already be more motivated.
- With randomization: Motivation should be distributed across groups → so outcome cannot be attributed to motivation.



Critique #1: Ethical Challenges of RCTs

1

Not all interventions can be randomized ethically --> medical treatments, social services, and education often make random assignment impossible or unjust.

2

Key questions arise: Who receives the benefit of the intervention? Is it acceptable to withhold a potentially effective treatment from the control group?

3

Scientific rigor vs. ethical feasibility: the stronger the experimental control, the greater the potential ethical cost to participants.

Ethical constraints do not eliminate the need for rigorous evaluation, they demand more creative research designs.

Critique #2: Practical and Implementation Challenges

- Even well-designed RCTs face serious real-world obstacles that can undermine the validity of estimated treatment effects.

Key Implementation Challenges

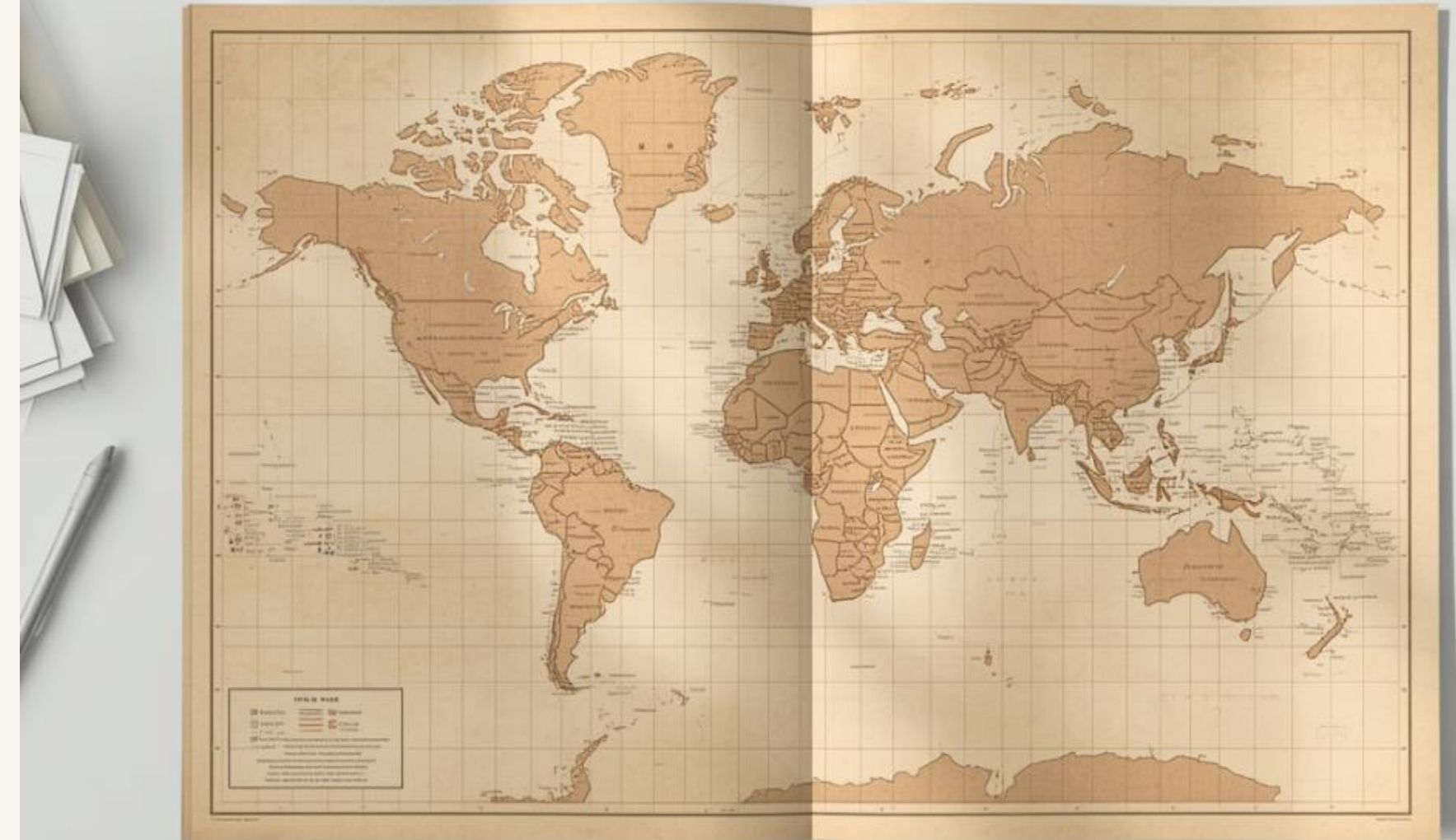
- Non-compliance: Participants assigned to treatment do not participate.
- Attrition: Participants are lost to follow-up over time.
- Contamination: Control group inadvertently receives similar services.
- Result: Estimated treatment effect may not reflect the true causal impact.



Critique #3: Generalizability and External Validity

TEAM A

- RCTs are often restricted to specific populations, places, and time periods, raising important questions about external validity.
- Findings may not generalize beyond the study's original context or community.
- A housing intervention that succeeds in one city may produce different outcomes elsewhere due to local economic conditions, demographics, or policy environments.
- Highly controlled experimental conditions can differ significantly from real-world settings.
- Selective participation further limits how broadly results can be applied
- **Strong internal validity does not guarantee that results travel well across contexts.**



Critique #4: Can RCTs Answer Every Research Question?

Factors That Cannot Be Randomized

- Some core social factors like poverty, race, economic conditions, neighborhood environment cannot be ethically or practically randomized.
- RCTs are simply not designed to study these structural influences on human outcomes.

Beyond Controlled Treatments

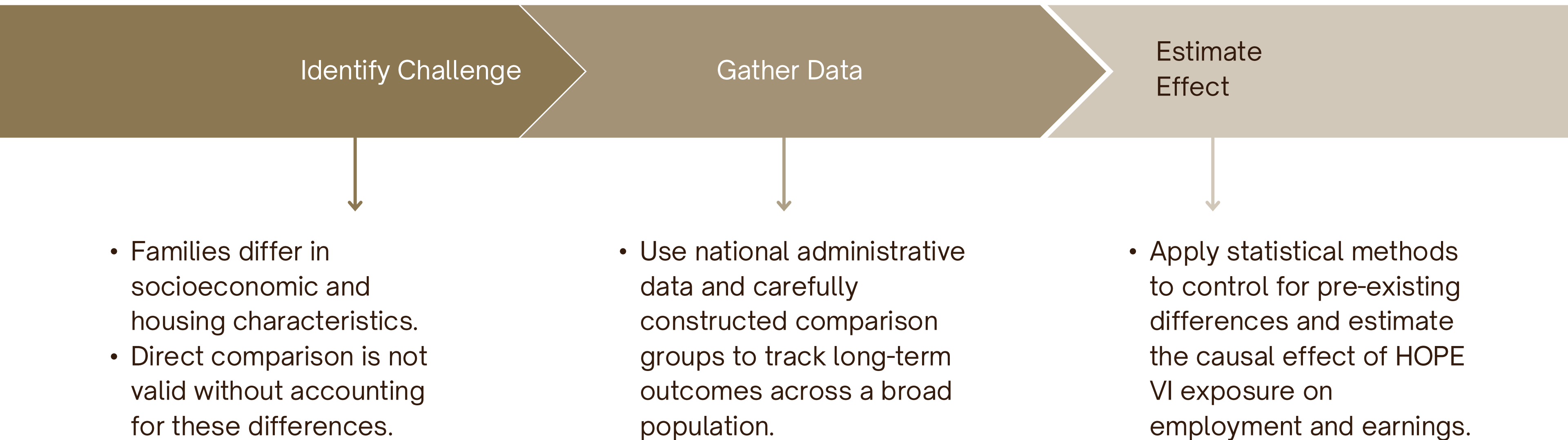
- Social science often studies environmental and systemic influences that unfold over time and across contexts.
- When RCTs cannot be applied, quasi-experimental designs become essential tools for understanding causality in these complex, real-world settings.

HOPE VI: The Challenge of Causal Inference

- The HOPE VI program demolished distressed public housing to reduce poverty concentration and improve long-term outcomes for residents.
- Researchers sought to study long-term labor market outcomes for children affected by HOPE VI demolitions, but a randomized controlled trial was not possible.
- Families were not randomly assigned to HOPE VI developments, making traditional experimental methods unfeasible.
- This raises a fundamental challenge: *How can we estimate the causal effect of a policy when assignment to treatment was not random?*



A Quasi-Experimental Alternative: Learning from Policy Variation



RCTs: Powerful, but Not Perfect

RCT Strengths

- Strong causal identification through randomization.
- Protection against selection bias and confounding variables.
- Clear treatment comparisons between groups.

When RCTs Work Best

- RCTs are the gold standard when randomization is ethical and practical.
- They provide the strongest evidence for causal inference in controlled settings.

Key Considerations

- Ethical constraints limit when randomization is feasible.
- Implementation problems such as non-compliance, attrition, and contamination can undermine results.

When Alternatives Are Needed

- Limited generalizability and inability to study many social phenomena make quasi-experimental methods essential.
- Research design must match the question.

Next Class

- Final Project Presentations
- Policy Report due Sunday 8/23 by 11:59pm

Questions?

CLASS 8: EXPERIMENTAL AND
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METHODS INTRODUCTION

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15-minute break